

Transformation in ETL

QUESTIONS:

Question 1: Define Data Transformation in ETL and explain why it is important.

Ans 1: Data Transformation in ETL is important because it helps to clean the data and with maximum accuracy missing values, null values may lead to the inaccuracy in the data and wrong insights that is why Data Transformation is important

For getting the right data.

Question 2 : List any four common activities involved in Data Cleaning.

Ans 2: Data Cleaning four common activities are:

1.Handling missing values: Missing values may led to the inconsistency into the data which may lead to the incorrect information of the data.

2.Removing duplicate data: Duplicate values takes up lot of spaces and it deletes the exact copy of the data present twice.

3.Correcting inconsistency data : Writing one thing into different formats which may lead to inconsistency into the data so we have to select one type of category and maintain the data on the behalf of that format.

4. Correcting Datatypes in data: Some data it must have given different datatypes of the values then we have to correct and write their correct format datatype .

Question 3 : What is the difference between Normalization and Standardization?

Ans 3: Normalization: It ranges the value between 0 and 1.

It range is fixed.

It effects the outliers.

Formula: $X - \min / \max - \min$

Standardization: It ranges the value Mean = 0 and SD = 1.

Range is not fixed.

It doesn't effect the outliers.

Formula : $X - \text{mean} / \text{SD}$

Question 4 : A dataset has missing values in the "Age" column. Suggest two techniques to handle this and explain when they should be used.

Ans 4: Two techniques to handle the missing values :

1. Mean/Median : to fill the values which are missing using mean median . median is majorly preferred because it will handle the outliers present in the data.

2. Remove the Rows : if only few are missing in the datasets we can remove them it doesn't affect the data .

The rows can be deleted.

Question 5 : Convert the following inconsistent "Gender" entries into a standardized format ("Male", "Female"): ["M", "male", "F", "Female", "MALE", "f"]

Ans 5: Given : we will choose the one category to use into the data .

("Male", "Female"): ["M", "male", "F", "Female", "MALE", "f"]

Either for male we choose : 'M' , MALE , 'male'

Either for female we choose : 'F' , 'FEMALE' , female.

Question 6 : What is One-Hot Encoding? Give an example with the categories: "Red, Blue, Green".

Ans 6: One hot encoding means converting the categorical data into numerical data into 0 and 1 form .

So that it can be used into data analysis and machine learning.

	Red	Blue	Green
Red	1	0	0
Blue	0	1	0
Green	0	0	1

Question 7 : Explain the difference between Data Integration and Data Mapping in ETL.

Ans7: Data Integration : It is used to combine the data from different different sources and integrate the data it main work is to bring the data into the system.

Example : SQL + Excel + APIs

Data Mapping : it is used to match the sources field with the targeted field. Also tells the relationship between the two columns.

Example : Customer_name , CUSTOMERNAME

Question 8 : Explain why Z-score Standardization is preferred over Min-Max Scaling when outliers exist.

Ans 8: Z-score Standardization is generally preferred when outliers exist because **Min-Max Scaling is highly affected by outliers.**

Min-Max scaling data converts the numbers into 0–1 range . if there are extreme outliers than it will compress it .

Z-score uses the **mean and standard deviation** to represent how far a value is from the mean.